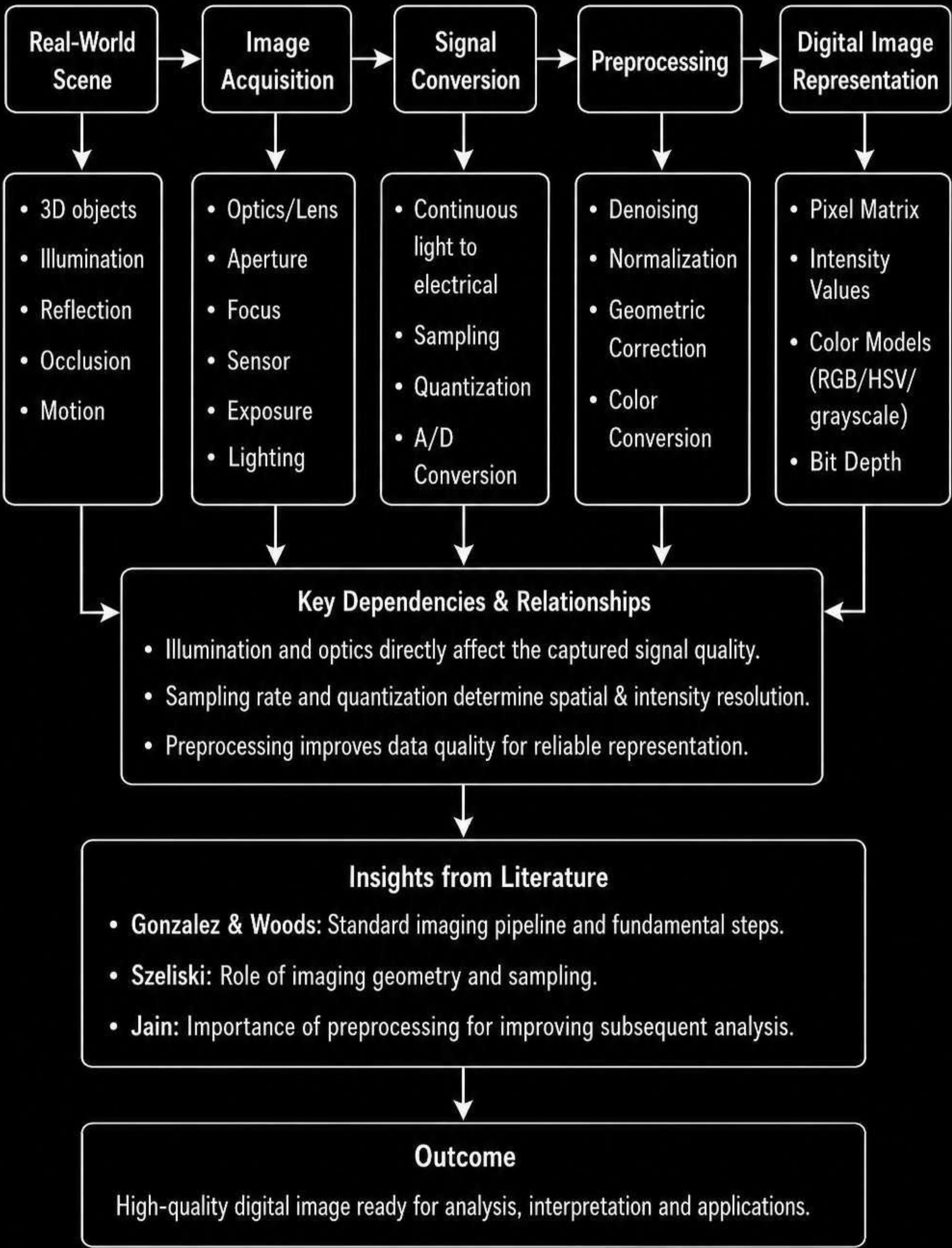
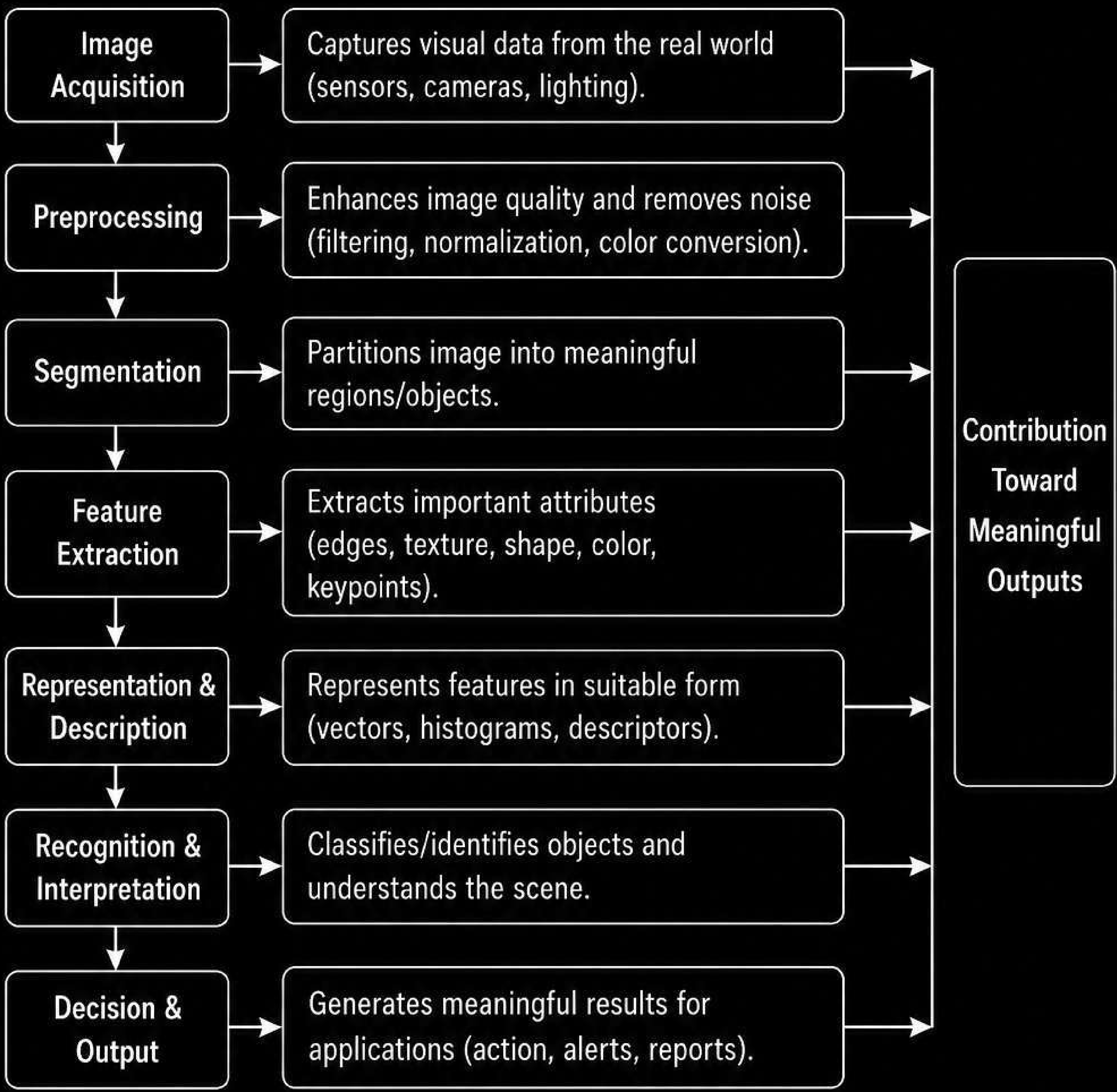
1. PIPELINE: REAL-WORLD SCENE CAPTURE TO DIGITAL IMAGE REPRESENTATION



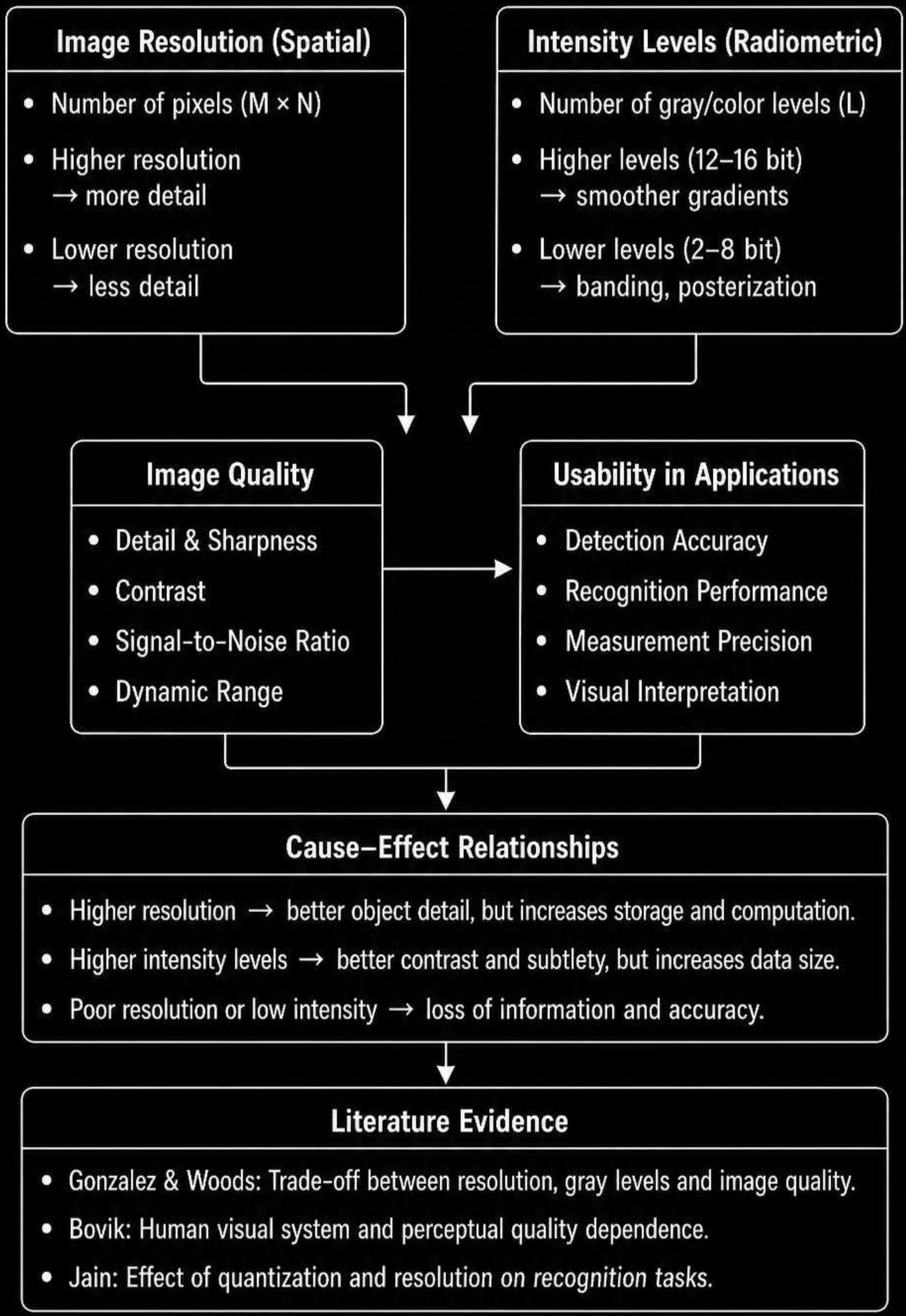
2. MAJOR STAGES OF VISUAL DATA HANDLING IN A COMPUTER VISION SYSTEM



Literature Connections

- **Forsyth & Ponce:** Vision pipeline from low-level processing to high-level understanding.
- **Szeliski:** Hierarchical processing and representation.
- **Bishop:** Pattern recognition & interpretation for decision making.

3. IMAGE RESOLUTION & INTENSITY LEVELS: IMPACT ON IMAGE QUALITY & USABILITY



4. IMAGE ACQUISITION CONDITIONS & THEIR IMPACT ON DOWNSTREAM PROCESSING AND APPLICATION PERFORMANCE

Acquisition Conditions → Effect on Captured Image → Impact on Downstream Processing → Effect on Application Performance

Acquisition Conditions

- Illumination (Lighting Conditions)
- Lens & Optics
- Sensor Characteristics
- Exposure Settings
- Environment
- Platform & Setup

Effect on Captured Image

- Brightness, shadows, reflections, highlights
- Focus, distortion, FOV
- Resolution, noise, bit depth, sensitivity
- Exposure time, gain, ISO, shutter speed
- Weather, background, occlusion, dynamic objects
- Camera position, angle, stability, vibration

Impact on Downstream Processing

- Low light → high noise
- Defocus → blur
- Distortion → geometric errors
- Low resolution → loss of details
- High noise → poor feature extraction
- Over/under exposure → saturation or dark image
- Occlusion/background clutter → false detections
- Vibration → motion blur, misalignment

Effect on Application Performance

- Accuracy (detection/recognition)
- Reliability (consistent results)
- Robustness (handling variations)
- Computational Cost (processing time)
- Overall System Performance

Key Relationships & Literature Support

- Acquisition conditions determine image quality, which directly influences all subsequent stages.
- Poor acquisition → error propagation → degraded final results.
- Seibert & Tees: Imaging conditions strongly affect vision system performance.
- Canny: Optimal imaging improves edge & feature extraction.
- Szeliski: Camera model & imaging parameters influence entire pipeline.

Summary

- Proper control and optimization of acquisition conditions → better image quality.
- Better image quality → improved accuracy, reliability and efficiency of vision applications.
- Ignoring acquisition factors → higher errors, more computation and poor real-world performance.

5. SYSTEM CAPABILITIES & THEIR INFLUENCE ON VISION-BASED APPLICATIONS

System Capabilities	What it Provides / Determines	Influence on Applications & Effectiveness	Examples of Influence
1. Sensing Capability	• Resolution, dynamic range, frame rate, spectral sensitivity • Determines the richness and quality of captured data	• Determines level of detail captured • Affects detection range, accuracy and reliability	High resolution + high dynamic range → fine inspection, medical imaging, surveillance
2. Processing Capability	• CPU/GPU power, parallelism, memory bandwidth • Determines speed and ability to handle complex tasks	• Enables real-time processing • Handles complex algorithms and large models efficiently	High processing power + high bandwidth → real-time detection, autonomous driving, AR/VR
3. Algorithmic Capability	• Feature extraction, learning ability, optimization, robustness • Determines intelligence and decision quality	• Improves accuracy, generalization and robustness • Adapts to diverse scenarios and conditions	Advanced algorithms + good learning ability → accurate recognition, anomaly detection, predictive analytics
4. Storage Capability	• Data capacity, read/write speed, retention, reliability • Determines amount of data that can be stored and accessed	• Supports long-term recording and analysis • Influences data throughput and system responsiveness	High capacity + fast storage → long-term video analytics, large dataset handling
5. Connectivity Capability	• Bandwidth, latency, reliability, real-time data transfer • Determines ability to communicate and share data	• Enables remote monitoring and control • Supports distributed and cloud-based operation	High bandwidth + low latency → IoT devices, edge computing, remote surveillance
6. Scalability & Integration	• Multi-sensor fusion, interoperability, extensibility • Determines ease of expansion and integration with other systems	• Allows system expansion without redesign • Enables integration in larger, complex systems	Scalable + interoperable → smart cities, industrial automation, multi-sensor systems

Key Relationships

- All capabilities are interdependent; improvement in one may require improvement in others.
- Balanced capabilities ensure optimal performance and resource utilization.
- Bottleneck in any capability limits overall system effectiveness and application success.

Literature Perspectives

- Khristian & Ness: System design must match application requirements and operating conditions.
- Goodfellow et al.: Deep learning performance depends on data quality, compute and model capability.
- Szeliski: Practical vision systems require balanced sensing, processing and algorithmic capabilities for reliable real-world performance.